SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 1.2: Chemicals of Life — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 1.2: Chemicals of Life
Driving Question	DRIVING QUESTION: Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies — and what does that tell us about what food is really made of?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Predict: What Is Food Really Made Of?**	Students examined familiar Kenyan foods (ugali, beans, sukuma wiki, milk, cooking oil) and recorded their initial predictions about what these foods are chemically made of. On the DQB, students posted their opening questions — e.g., 'Why does eating only ugali make you weak?' and 'What is cooking oil made of?' Teacher should note which misconceptions surface most frequently (common one: students equate food with energy only, missing structural roles). Students also sketched their INITIAL MODEL of how food becomes 'part of the body.'	Students learned that food cannot be described simply as 'energy' — it is a mixture of distinct chemical substances, each with a specific role in the body. They were introduced to the anchoring phenomenon: two children in Kisumu eating the same quantity of food but developing differently, which cannot be explained by amount alone. This lesson establishes that the TYPE and COMPOSITION of food matters. Pedagogical note: Emphasise that 'predict' is not about being right — it is about activating prior knowledge and creating cognitive hooks for new learning.	Students began explaining the phenomenon by recognising that identical quantities of food are not chemically identical — a plate of ugali and a plate of githeri differ in chemical composition, not just taste. This is the first wedge into the driving question: the two children must be eating foods with different chemical makeups. Teacher should cold-call 3–4 students to share their initial model sketches and explicitly affirm that all predictions are valid starting points. Use WAIT TIME of at least 5 seconds before accepting answers.
Lesson 2 **What Is Food Really Made Of? Introducing the Six Chemicals of Life**	Students conducted food tests (Benedict's test for reducing sugars using maize porridge, iodine test for starch using ugali, Biuret test for proteins using milk and bean soup, and grease-spot test for lipids using groundnuts). They recorded colour changes and results in data tables. On the DQB, students added new questions arising from the tests — e.g., 'Why does ugali turn blue-black with iodine but milk does not?' Teacher should circulate during practicals	Learners established that all foods are composed of six major chemicals of life: carbohydrates, proteins, lipids, nucleic acids, vitamins/minerals, and water. Each chemical has a distinct identity, structure, and function in the body. Key vocabulary introduced: monomer, polymer, macromolecule, micronutrient. Students learned that the food tests are tools for detecting specific chemicals — the tests do not change the food, they reveal what is	Students used their food test results to begin building a chemical profile of common Kenyan foods. They explained that the two children in Kisumu are likely consuming different chemical profiles — one may be getting more proteins (supporting muscle and tissue growth) while the other gets more carbohydrates (providing energy but less structural building material). Teacher should ask: 'If both children eat the same weight of food but one eats mostly ugali and

	and prompt students with: 'What does this colour change tell you about what is inside this food?'	already there. Pedagogical note: Reinforce that nucleic acids are rarely tested for in a school lab but are critically important — connect to DNA and inheritance.	the other eats githeri — which chemicals differ, and what body difference would you predict?' Allow pair-share before whole-class discussion. Students revise their INITIAL MODEL to include the six chemical categories.
Lesson 3 **Observe (Part 2): Lipids and Vitamins — Structure, Function, and Food Tests**	Students performed the emulsion test for lipids (using cooking oil and water with ethanol) and analysed case studies of Kenyan children with vitamin deficiency diseases — e.g., night blindness (Vitamin A deficiency) in fishing communities near Lake Victoria, and scurvy symptoms in areas with low fruit intake. Students watched or read about how fat-soluble vitamins (A, D, E, K) require dietary fat to be absorbed. On the DQB, students added: 'Why do you need fat to absorb some vitamins?' and 'Can eating too much fat be harmful?' Teacher should highlight the real-world public health relevance of these deficiencies in Kenya.	Learners established that lipids are composed of glycerol and fatty acids joined by ester bonds, and that they serve three critical roles: long-term energy storage (more than twice the energy per gram of carbohydrates), structural components of cell membranes (phospholipid bilayer), and carriers for fat-soluble vitamins. Vitamins and minerals, though needed in tiny amounts (micronutrients), are essential for enzyme function, immune response, and bone health. Deficiency — not just deficiency in quantity of food — causes disease. Pedagogical note: Use the Lake Victoria fishing community context to make vitamin A deficiency tangible and locally relevant.	Students extended their explanation of the phenomenon: the two children in Kisumu may have different body compositions not just because of carbohydrate or protein differences, but because one child's diet lacks sufficient fat or micronutrients, affecting cell membrane integrity, vitamin absorption, and metabolic regulation. Teacher should guide students to update their MODELS to include lipids and vitamins as distinct chemical categories with structural and regulatory (not just energy) roles. Ask: 'Could a child become ill even if they eat enough ugali every day? What would be missing?' Cold-call at least 3 students.
Lesson 4 **Building Life's Molecules: Structure of Carbohydrates and Proteins**	Students used physical or drawn models (e.g., paper chain links representing glucose monomers) to build mono-, di-, and polysaccharide chains, observing how glycosidic bonds link units and how condensation reactions release water. They similarly modelled amino acid chains (peptide bonds) to form dipeptides and polypeptides. Students examined how the sequence of amino acids determines the shape and function of a protein. On the DQB, students posted: 'Does it matter which order the amino acids are in?' and 'How does the body break these chains apart?' Teacher should connect condensation (building) to hydrolysis (breaking down during digestion) as complementary processes.	Carbohydrates are built from monosaccharide monomers (e.g., glucose) linked by glycosidic bonds via condensation reactions to form disaccharides (e.g., sucrose in chai, maltose in fermented uji) and polysaccharides (e.g., starch in ugali, glycogen stored in the liver). Proteins are built from amino acid monomers linked by peptide bonds — the specific sequence of amino acids (determined by DNA) gives each protein its unique 3D shape and function. Key insight: structure determines function at the molecular level. Pedagogical note: Students often confuse condensation (synthesis, releases water) with hydrolysis (digestion, uses water) — use a clear mnemonic or diagram to distinguish them permanently.	Students connected molecular structure to the phenomenon: the two children differ in body composition partly because their diets supply different amino acid profiles. If one child's diet lacks essential amino acids (those the body cannot synthesise), their body cannot build the proteins needed for growth, repair, and enzymes — even if total food quantity is equal. Teacher should ask: 'Maize (ugali) lacks the amino acid lysine. If a child eats only ugali, what protein-building problem does their body face?' Students revise their MODELS to show monomers → polymers and label the bonds involved. Collect models for formative assessment after this lesson.
Lesson 5 **Nucleic Acids and Water: The Molecules That Carry the Blueprint**	Students examined diagrams of DNA and RNA structure, identifying the nucleotide monomer (sugar + phosphate + nitrogenous base), the double helix, and the base-	Nucleic acids (DNA and RNA) are polymers of nucleotide monomers. DNA stores genetic information in the sequence of its bases; this sequence ultimately codes for	Students now have a near-complete explanation of the phenomenon: the two children in Kisumu differ because (1) their diets may contain different chemical

and Sustain Life	pairing rules (A-T, G-C in DNA). They connected nucleic acid structure to its function: DNA stores the genetic instructions for which proteins to build. Students also investigated the unique properties of water (cohesion, high specific heat capacity, solvent properties) through simple observations — e.g., why ugali sticks together, why sweat cools the body. On the DQB, students added: 'How does DNA tell the body which proteins to make?' and 'Why do we die faster from thirst than hunger?' Teacher should note this lesson bridges chemistry and genetics — a powerful conceptual connection.	the amino acid sequence of every protein in the body. RNA carries this information from the nucleus to ribosomes. Water, though not a macromolecule, is the most abundant chemical in cells and is essential as a solvent, reactant (in hydrolysis), temperature regulator, and transport medium. Pedagogical note: Help students see the full chain — DNA sequence → protein sequence → protein shape → protein function → body structure and metabolism. This is the molecular explanation for why two children with different genetics build different bodies even on similar diets.	profiles, AND (2) their DNA codes for different proteins, meaning their bodies use the same food chemicals differently. Teacher should facilitate a class discussion: 'Is the difference between the two children caused by food, genes, or both? Use evidence from Lessons 1–5.' Students update their CUMULATIVE MODELS to include nucleic acids as the blueprint layer above all other chemicals. Remind students that water connects everything — it is the medium in which all chemical reactions of life occur.
Lesson 6 **Final Explanation: Pulling It All Together — What Food Is Really Made Of**	Students reviewed all entries in their Summary Tables, compared their INITIAL MODEL (Lesson 1) with their FINAL MODEL (revised through Lessons 2–5), and completed the DQB by answering or flagging remaining questions. They also reviewed their food test data, molecular diagrams, and case studies from across the unit. Teacher should allocate time for students to share their final models in pairs or small groups before the whole-class synthesis. This lesson serves as both formative consolidation and preparation for the final written explanation task.	Students consolidated understanding of all six chemicals of life — carbohydrates (energy and structure), proteins (structure, enzymes, transport), lipids (long-term energy, membranes, vitamin transport), nucleic acids (genetic blueprint for protein synthesis), vitamins and minerals (regulatory micronutrients), and water (universal solvent and reactant). They understand that food quality (chemical composition) determines body outcomes, not food quantity alone. They can connect molecular structure to cellular function to whole-body development — the full explanatory chain required by the unit driving question. Pedagogical note: Use this lesson to explicitly celebrate the growth in student thinking — show side-by-side Lesson 1 vs. Lesson 6 models to make learning visible.	Students constructed their FINAL EXPLANATION of the driving question: The two children in Kisumu end up with different bodies because food is not just 'food' — it is a mixture of specific chemicals (carbohydrates, proteins, lipids, nucleic acids, vitamins, minerals, and water), each with distinct molecular structures and functions. If their diets differ in chemical composition — e.g., one gets adequate protein with all essential amino acids while the other subsists on carbohydrate-heavy meals — their bodies receive different molecular raw materials. Additionally, each child's DNA provides a unique blueprint that directs how these chemicals are assembled into proteins and structures, meaning genetic differences further amplify dietary differences. Teacher should cold-call 4–5 students to deliver one sentence each of the final explanation, building it collaboratively on the board. Collect Summary Tables for marking.